Shray Komal

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Education

University of Houston
Master of Science in Computational Health Informatics- Data Science, GPA: 3.83/4.0 May 2025
Bachelor of Science in Biotechnology – Bioinformatics, GPA: 3.52/4.0 May 2024
Relevant Coursework: Data Science, Genomics, Machine Learning, Deep Learning, Large Language Models, Generative AI
Certifications: AWS Certified AI Practitioner, Google Data Analytics Certificate, Innovation Engineering Blue Belt
Academic Awards: Dean's List x 6, Academic Excellence Award Scholarship, Magna Cum Laude

Skills

Languages and Libraries: Python, Microsoft SQL server, NLKT, spaCy, R, MATLAB, Pandas, Tensorflow, PyTorch, R shiny, Prophet, XGBoost
Softwares and IDEs: AWS Sagemaker, PowerBI, Linux, Ubuntu, Databricks, Hadoop, GIT, Excel, SAP Datasphere and Analytics Cloud, Nextflow

Experience

Perry Homes Houston, Texas
Data Scientist Intern March 2025 – Present

- Developed and deployed machine learning models using libraries such as **Prophet, XGBoost, and Statsmodels** to forecast construction inventory needs which lead to a **121% increase in profit margins** by reducing material overstock and improving supply planning
- Automated legacy file organization systems using **Power Automate** and SharePoint workflows, resulting in weekly time **savings of 44 hours** for staff across departments.
- Engineered anomaly detection models in Python using statistical thresholds and tree-based classifiers to flag irregular supply-demand patterns, reducing inventory planning errors by **29%**

Trident 1 Remote
Business Data Science Intern October 2024 – March 2025

- Developed and implemented **Python**-based **NLP** models using **BERT** using **AWS Sagemaker** to automate document processing and successfully extracting data from 50+ client documents with **99% accuracy**, reducing manual processing time by **25 hour weekly**
- Orchestrated enterprise-wide ecommerce integration using **Jira** to standardize processes across 6 platforms by centralizing data management for 50+ clients while achieving 65% faster reporting and enhancing data accuracy from by **17%**

Hilcorp Energy Houston, Texas
Data Analyst Co-op November 2022 – October 2024

- Utilized **SAP Datasphere** and **SQL** to design data models, improving the data usability by 45% while across all 7 departments
- Developed **Power BI** reports and SAP Analytics Cloud dashboards, driving a 32% increase in operational visibility
- Automated a **Python** script utilizing **Git** for version control for the extraction of relevant data from CSV files and convert them into required formats for further statistical analysis of oil production data using Pandas

Simulation Engineering Lab Remote
Data Science Intern July 2023 – September 2023

- Built a deep learning model using ReLU-based **Convolutional Neural Networks** that achieved 98.2% accuracy for kidney cancer detection by optimizing hyperparameters led to F1 Score of 86% and ROC AUC of 93%
- Processed CT scan medical images from company data using augmentation techniques in **TensorFlow**, and performed exploration analysis to identify data quality issues, improve model performance, visualize insights in a Plotly dashboard for stakeholders

Academic Capstone Project

Breast Cancer Machine Learning Model Houston, Texas
Bioinformatics Data Scientist August 2023 – August 2025

- Preprocessed TCGA methylation data and applied dimensionality reduction to optimize **artificial neural network** (deep learning model) input features to capture complex relationships in methylation data to potentially identify highly significant biomarker and lead to 92.1% accuracy
- Established REST API to retrieve data, built an artificial neural network (ANN) using **PyTorch**, **GIT** for version control for breast cancer stage prediction, and **Nextflow** for management of workflow pipeline

Target Success Predictor Dashboard Houston, Texas
Bioinformatics Analyst January 2025 – May 2025

- Built a machine learning model Random Forest with ROC AUC of 0.83 to predict Phase III trial success using structured clinical data and features derived from a Neo4j knowledge graph
- Developed an R Shiny dashboard to display predictions, success metrics by therapeutic area, and live Neo4j queries that cut research lookup time by 70% for analysts

Leadership Experience

Future IT Professionals Houston, Texas
Operations Officer August 2022 – Present

- Managed 4 coordinators and applied foresight principles to increase the number of company information sessions by 83% by implementing weekly operational plans and database management using **AWS RDS** and Google sheets
- Overlooked organizational recruitment, increasing membership growth by 36% from the previous academic year